

Aggregate Query Engine — Rebuilt Architecture

Handover for Manual QA | Request for Test Queries

Hi team,

We have completed a substantial rebuild of the aggregate query engine — the component that answers “how many...” questions across cases. Before we call it done, we need your judgement on it. Below is what changed, the full evidence from our testing, and what we need from you.

1. What we rebuilt, and why it mattered

The previous engine matched questions against a fixed set of pre-written queries. It had a specific and dangerous failure mode: **when it did not fully understand a question, it answered a simpler one instead — and reported full confidence.**

The new architecture interprets each question into an explicit structured query plan, validates that plan against what the database actually contains, and only then runs it. If the question asks for something the data cannot express, it refuses and states why, rather than quietly answering a different question.

2. Test coverage

37 distinct questions, 271 executions. Most questions were run 3 times, and two were run 20 times each to measure repeatability. Every figure in this document was verified against the live database.

Test set	Questions	Executions	Purpose
Blind question corpus (before & after)	25	150	broad coverage; the primary before/after comparison
Semantic adjacency	5	15	do near-identical questions stay distinct?
Blind category sweep	7	21	counting, relational, temporal, multi-hop, unsupported
Repeatability (20 runs each)	2	80	same question, same answer?
Live end-to-end via /api/chat	5	5	real login, real HTTP, real user path

Automated tests: 386 covering the redesigned architecture — all passing. The full application suite (3,959 tests) also passes with no regressions.

3. Headline results

Measure (25-question corpus, 3 runs each)	Before	After
Questions giving inconsistent answers	6	1
Crashes	1	0
Silently wrong answers	5	0
Repeatability of one complex question	1 / 20	20 / 20

4. Where the architecture was previously wrong — and is now correct

These are the cases that justified the rebuild. Every “old answer” was returned with full confidence and no warning.

Question	Old engine	New engine	Verified truth
How many people have been accused in incidents?	206	92	92 — 206 was everyone <i>involved</i> , not the accused
How many persons have a risk score above 80 ?	208	refused	field does not exist; 208 was every person in the database

Question	Old engine	New engine	Verified truth
How many licensed weapons are in evidence?	0 / 32 / 32	refused	0 licensed — 30 are unlicensed, 2 unknown; 32 was all weapons
How many officers assigned, and how many cases have officers?	76	refused	two questions; the second half was silently dropped
What is the earliest incident date ?	crash	2024-09-14	engine could not handle a non-numeric answer
How many accused persons are still under investigation?	55 / refused	refused	55 counted an unrelated population; needs two filters we cannot express
How many distinct CNIC numbers , and how many persons have one?	194 / 208 / refused	refused	gave three different answers on three runs

5. Questions answered correctly and repeatably

Every answer below was identical on all three runs and confirmed against the database.

Question	Answer	Confirmed
How many cases are recorded in the system?	73	73
How many weapons are in the evidence records?	32	32
How many police stations are there?	19	19
How many unlicensed weapons are recorded?	30	30
How many documents are FIR records?	73	73
How many female persons are in the records?	34	34
How many cases have a weapon linked to them?	32	32
How many people are connected to cases?	208	208
How many distinct persons appear across all cases?	208	208 — <i>previously refused outright</i>
How many people have been accused in incidents?	92	92 — <i>previously answered 206</i>
What is the date of the earliest recorded incident?	2024-09-14	<i>previously crashed</i>

6. Generalisation — questions the engine had never seen

These were written after the rebuild was finished, to test whether it generalises or was merely tuned to our own examples.

Near-identical questions must stay distinct

Question	Answer	Assessment
How many accused persons are linked to cases?	92	correct — distinct from all three below
How many witnesses participated?	37	correct
How many victims are recorded?	9	correct
How many people are involved in cases?	208	correct — the broad question, correctly broad
How many suspects have confirmed arrest status?	refused	correct: needs two filters on one relationship — a known limit

This is the central result. The old engine collapsed all four of the first questions into one broad count. The new engine separates them, because role information lives on the *relationship* between a person and an incident — something it now discovers automatically.

Category sweep

Category	Question	Result
Counting	How many cases exist?	73 — correct
Relationship	How many people have role witness?	37 — correct
Adversarial	How many unicorn weapons were seized?	0, flagged as too few to be meaningful
Unsupported	How many people are left-handed ?	refused — no such field; never invented a number
Temporal	How many incidents occurred after 2024?	refused — ambiguous path, reason stated
Multi-hop	How many people are connected through these cases?	refused — exceeds supported hop limit

7. Live end-to-end through the real application

Five requests through **/api/chat** with a real login and real session — not internal test calls. All returned HTTP 200 with zero exceptions.

Question asked in chat	Response
How many cases are registered?	73 — correct
How many accused persons are linked to cases?	92 — correct, with an honest note that independent verification could not confirm it
How many witnesses are recorded?	37 — correct
How many accused persons have a criminal risk score above 80?	refused — “the field ‘criminal_risk_score’ does not exist”. The old engine answered 208 .
How many people are left handed?	refused — out of scope, no number invented

8. Two things to know before testing

A refusal is often the correct answer. The engine now declines questions it cannot answer precisely. “*The field X does not exist*” is it working, not failing. Please report unclear wording — but a refusal itself is not a bug.

“Verification did not confirm this figure” is honesty, not failure. For some question types our independent cross-checker cannot express the same query, so we say so rather than implying more confidence than we have.

One known cosmetic issue: the response text sometimes hedges (“counted 37 entities”) even when the number is correct. Please flag the phrasing, not the figure.

9. What we need from you

Please send the aggregate queries you want tested — the ones that matter operationally, not just easy ones. Most valuable:

- Questions involving **roles** (accused / witness / victim / complainant)
- Questions spanning **relationships** (persons → incidents → cases → stations)
- **Date-ranged** questions
- **Multi-part** questions
- Questions you **suspect might break it** — genuinely the most useful category

For each, tell us what you would consider a correct answer, or say you do not know — both are useful. We will run them and report back with the answer *and* its full reasoning trail, so you can judge whether the system understood the question rather than just whether the number looks reasonable.

20–40 queries makes a good first batch.

Thanks — your testing is what tells us whether this holds up against real investigative questions rather than ours.